
How to Create a Docker Container and Store the Image in AWS ECR

Prerequisites

- Docker installed and running.
 - AWS CLI configured with permissions.
 - An ECR repository created (see previous steps or create one now).
 - AWS account ID and region known.
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Step 1: Write a Dockerfile

Create a directory for your app and inside it, create a file named **Dockerfile**. Here's an example Dockerfile for a simple Node.js app:

```
# Use official Node.js runtime as a parent image
FROM node:18-alpine
```

```
# Set working directory inside the container
WORKDIR /app
```

```
# Copy package.json and package-lock.json
COPY package*.json ./
```

```
# Install dependencies
RUN npm install
```

```
# Copy the rest of your app's source code
COPY . .
```

```
# Expose port 3000
```

EXPOSE 3000

Command to run your app
CMD ["node", "index.js"]

Make sure you have your Node.js app files in the same directory (`package.json`, `index.js`, etc.).

Step 2: Build the Docker Image Locally

Navigate to the directory containing your Dockerfile:

```
docker build -t my-node-app .
```

Step 3: Create an ECR Repository (if you haven't already)

```
aws ecr create-repository --repository-name my-node-app --region us-west-2
```

Step 4: Authenticate Docker to Your ECR Registry

```
aws ecr get-login-password --region us-west-2 | docker login --username AWS --password-stdin  
<aws_account_id>.dkr.ecr.us-west-2.amazonaws.com
```

Replace `<aws_account_id>` with your AWS account number.

Step 5: Tag the Docker Image for ECR

```
docker tag my-node-app:latest  
<aws_account_id>.dkr.ecr.us-west-2.amazonaws.com/my-node-app:latest
```

Step 6: Push the Docker Image to ECR

```
docker push <aws_account_id>.dkr.ecr.us-west-2.amazonaws.com/my-node-app:latest
```

Step 7: Verify the Image in ECR

Go to the AWS Console → ECR → Your repository (**my-node-app**) and verify that your image appears there.

Summary Table

Step	Command/Action
Write Dockerfile	Create Dockerfile with your app setup
Build Image	<code>docker build -t my-node-app .</code>
Create ECR Repo	<code>aws ecr create-repository --repository-name my-node-app --region us-west-2</code>
Authenticate Docker to ECR	<code>`aws ecr get-login-password ...</code>
Tag Image	<code>docker tag my-node-app:latest <aws_account_id>.dkr.ecr.us-west-2.amazonaws.com/my-node-app:latest</code>
Push Image	<code>docker push <aws_account_id>.dkr.ecr.us-west-2.amazonaws.com/my-node-app:latest</code>
